

E077

Introduction to AI Engineering

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A regional insurer with a new downtown office asks three staff members for help with claims. One will study last year's claims to explain why costs rose. One will test a brand-new neural network design and write a paper on it. One will package an existing fraud model as a service, set up monitoring, and keep it running at 99.9% uptime. The team lead drinks a lot of coffee. Which person is doing AI engineering work?

- A. The person packaging the fraud model as a monitored service.
- B. The person explaining why claim costs rose.
- C. All three, because every role uses data and code.
- D. The person testing a new neural network design for a paper.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

An engineer at a logistics startup times a matching routine that compares every shipment with every other shipment, an $O(n^2)$ process. It takes 2 seconds on 10,000 shipments on her 16 GB laptop, which she bought last spring. About how long should it take on 30,000 shipments?

- A. About 18 seconds.
- B. It cannot be estimated without knowing the laptop's memory.
- C. About 2 seconds, because the code has not changed.
- D. About 6 seconds.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A linear $O(n)$ cleaning job takes 4 seconds on 1 million rows. The time budget is 12 seconds. About how many rows can the job handle within budget?

- A. About 3 million rows.
- B. About 1.7 million rows.
- C. About 12 million rows.
- D. About 333,000 rows.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A team at a mid-sized retailer builds a demand model. They try 40 settings, pick the one that scores best on the test set, and report that test score, 94%, to executives in a glossy slide deck. The data covers five years of sales. What is the main problem with the reported 94%?

- A. There is no problem, because 94% is above the usual 90% bar.
- B. It is optimistic, because the test set was used for tuning and no longer measures unseen data.
- C. It is optimistic, because 40 settings is too few to test.
- D. It is too low, because five years of data should give higher accuracy.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A model scored 91% offline but only 70% in production. The team learns the scaler was fitted on the full dataset before it was split. What does this suggest?

- A. Scaling should never be used in machine learning.
- B. The model is underfitting and needs more layers.
- C. Data leakage inflated the offline score.
- D. Production users are entering bad data, so the model is fine.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A clinic's no-show predictor, built by a two-person team working remotely, scores 99% accuracy on training data and 68% on validation data. The clinic's parking lot was recently repaved. What is the most likely issue and first fix?

- A. Underfitting; add more layers and train longer.
- B. Overfitting; add regularization or simplify the model.
- C. No issue; 99% accuracy proves the model works.
- D. Overfitting; train for more epochs.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A model scores 61% on training data and 60% on validation data for a task where simple rules already reach 85%. What best describes the model?

- A.** Well fit, because training and validation match.
- B.** Overfitting, because the scores are close.
- C.** Data leakage, because the scores are nearly equal.
- D.** Underfitting; it is too simple to capture the pattern.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A bank's fraud model, trained on GPUs rented for \$2 an hour, flagged 200 transactions last month. Of those, 150 were real fraud. Auditors later found 250 fraudulent transactions in total that month. What is the model's precision?

- A.** 0.60
- B.** 0.80
- C.** 0.25
- D.** 0.75

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A screening model has recall of 0.80. There were 400 actual positive cases. How many true positives did it catch?

- A.** 80
- B.** 500
- C.** 320
- D.** 400

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A chatbot team at a travel company, located near a busy airport, switches its model server to batch 16 requests at a time. Requests per second rise from 40 to 110, but each user now waits 450 ms instead of 180 ms. The team repainted its office in blue. What happened?

- A.** Both improved, because the server does more work.
- B.** Throughput improved while latency got worse.
- C.** Both got worse, because users wait longer.
- D.** Latency improved while throughput got worse.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A food-delivery firm's ETA model, written in a popular open-source framework, was accurate at launch. Six months later a new neighborhood opened and many new addresses appear. Errors grow, but no alert fired. The model file has a 4.2 GB size. What should the team have had in place?

- A.** A new test set built from launch-day data.
- B.** A larger model file to hold more addresses.
- C.** Nothing; a model that passed testing stays accurate.
- D.** Monitoring for data and performance drift with alerts and a fallback.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A lender's loan-approval model, built by a highly rated vendor based overseas, reaches 92% accuracy overall. The vendor removed race and gender columns from the inputs. The lender's headquarters won a design award. What should the lender do before using it for decisions?

- A.** Deploy it, because 92% overall accuracy is high.
- B.** Ask the vendor for a lower price, because it is based overseas.
- C.** Test performance and error rates across groups and set up human review for denials.
- D.** Deploy it; removing race and gender guarantees fairness.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A regional insurer ~~with a new downtown office~~ asks three staff members for help with claims. One will study last year's claims to explain why costs rose. One will test a brand-new neural network design and write a paper on it. One will package an existing fraud model as a service, set up monitoring, and keep it running at 99.9% uptime. ~~The team lead drinks a lot of coffee.~~ Which person is doing AI engineering work?

Target word: which role is AI engineering **Decides:** Deploying, monitoring, and operating a model as a reliable service.

A	Correct	Building and operating a dependable production system is the core of AI engineering.
B	Adjacent category	Answering a business question from data is data science.
C	Absolute language	Shared tools do not make the roles the same; the deliverable differs.
D	Confusion set	Inventing and publishing new methods is research.

2. Answer A Concept 2

An engineer at a logistics startup times a matching routine that compares every shipment with every other shipment, an $O(n^2)$ process. It takes 2 seconds on 10,000 shipments ~~on her 16 GB laptop, which she bought last spring.~~ About how long should it take on 30,000 shipments?

Target word: run time at $3\times$ input **Decides:** $O(n^2)$: tripling n multiplies time by $3^2 = 9$.

A	Correct	$2\text{ s} \times 3^2 = 18\text{ s}$.
B	Camouflage	Hardware sets the base time; the growth rate comes from the algorithm.
C	Dropped element	Ignores the effect of input size on run time.
D	Adjacent category	Applies linear $O(n)$ growth to a quadratic routine.

3. Answer A Concept 2 · transfer

A linear $O(n)$ cleaning job takes 4 seconds on 1 million rows. The time budget is 12 seconds. About how many rows can the job handle within budget?

Target word: solve for input size **Decides:** Linear: time and rows scale by the same factor ($12 / 4 = 3$).

A	Correct	$3\times$ the time allows $3\times$ the rows for an $O(n)$ job.
B	Adjacent category	Uses the square root, as if the job were $O(n^2)$.
C	Wrong stem number	Treats the 12-second budget as a row count.
D	Inverted operation	Divides by 3 instead of multiplying.

4. Answer B Concept 3

A team at a mid-sized retailer builds a demand model. They try 40 settings, pick the one that scores best on the test set, and report that test score, 94%, to executives ~~in a glossy slide deck. The data covers five years of sales.~~ What is the main problem with the reported 94%?

Target word: why the score is untrustworthy **Decides:** The test set was used to choose the model, so it is no longer unseen.

A	Absolute language	A high number measured the wrong way is still unreliable.
B	Correct	Selecting on the test set leaks it into the model choice; a separate validation set should be used for tuning.
C	Inverted operation	Trying more settings on the test set would make the leakage worse, not better.
D	Camouflage	The data span does not decide whether the evaluation is valid.

5. Answer C Concept 3 · transfer

A model scored 91% offline but only 70% in production. The team learns the scaler was fitted on the full dataset before it was split. What does this suggest?

Target word: diagnose the gap from its cause **Decides:** Preprocessing fitted on all data leaks test information into training.

A	Absolute language	Scaling is fine when fitted on training data only.
B	Adjacent category	Underfitting shows low scores everywhere, not a high offline score.
C	Correct	Fitting on all rows lets test statistics shape training; the offline score is optimistic.
D	Dropped element	Ignores the leak the team found.

6. Answer B Concept 4

A clinic's no-show predictor, ~~built by a two-person team working remotely,~~ scores 99% accuracy on training data and 68% on validation data. ~~The clinic's parking lot was recently repaved.~~ What is the most likely issue and first fix?

Target word: diagnose and remedy **Decides:** Large gap between training and validation scores = overfitting.

A	Confusion set	Underfitting shows low training scores too; this fix would worsen overfitting.
B	Correct	The model memorized training data; reducing complexity or adding regularization narrows the gap.
C	Absolute language	Training accuracy alone says nothing about new data.
D	Inverted operation	More training usually deepens memorization.

7. Answer D Concept 4 · transfer

A model scores 61% on training data and 60% on validation data for a task where simple rules already reach 85%. What best describes the model?

Target word: classify from low, close scores **Decides:** Both scores are low and close together.

A	Dropped element	Ignores that both scores are far below a simple baseline.
B	Inverted operation	Overfitting shows a large gap, not close scores.
C	Adjacent category	Leakage inflates scores; it does not produce uniformly low ones.
D	Correct	Low scores on both sets signal high bias.

8. Answer D Concept 5

A bank's fraud model, ~~trained on GPUs rented for \$2 an hour~~, flagged 200 transactions last month. Of those, 150 were real fraud. Auditors later found 250 fraudulent transactions in total that month. What is the model's precision?

Target word: precision **Decides:** Precision = true positives / all flagged = 150 / 200.

A	Confusion set	150 / 250 is recall, not precision.
B	Wrong stem number	200 / 250 divides flags by actual fraud; neither is the true-positive count.
C	Inverted operation	That is the false-positive share of flags (50 / 200), not precision.
D	Correct	150 / 200 = 0.75.

9. Answer C Concept 5 • transfer

A screening model has recall of 0.80. There were 400 actual positive cases. How many true positives did it catch?

Target word: solve for true positives **Decides:** Recall = TP / actual positives, so TP = 0.80 × 400.

A	Inverted operation	That is the number missed (false negatives).
B	Adjacent category	400 / 0.80 treats 400 as true positives and solves for a total.
C	Correct	0.80 × 400 = 320.
D	Wrong stem number	The count of actual positives, not the number caught.

10. Answer B Concept 6

A chatbot team at a travel company, ~~located near a busy airport~~, switches its model server to batch 16 requests at a time. Requests per second rise from 40 to 110, but each user now waits 450 ms instead of 180 ms. ~~The team repainted its office in blue.~~ What happened?

Target word: effect on latency vs. throughput **Decides:** Requests per second rose; time per request rose.

A	Dropped element	Ignores that wait time rose from 180 ms to 450 ms.
B	Correct	Batching completes more requests per second but each request waits longer.
C	Confusion set	Treats latency and throughput as the same measure.
D	Inverted operation	Swaps the two measures.

11. Answer D Concept 7

A food-delivery firm's ETA model, ~~written in a popular open source framework~~, was accurate at launch. Six months later a new neighborhood opened and many new addresses appear. Errors grow, but no alert fired. ~~The model file has a 4.2 GB size.~~ What should the team have had in place?

Target word: missing reliability control **Decides:** Input data changed over time without detection.

A	Dropped element	Old data would not reveal the new neighborhood's shift.
B	Camouflage	File size does not detect or fix drift.
C	Absolute language	Real-world data shifts; tested models degrade.
D	Correct	Drift monitoring catches changing inputs; a fallback keeps service safe while the model is retrained.

12. Answer C Concept 8

A lender's loan-approval model, ~~built by a highly-rated vendor based overseas,~~ reaches 92% accuracy overall. The vendor removed race and gender columns from the inputs. ~~The lender's headquarters won a design award.~~ What should the lender do before using it for decisions?

Target word: responsible-use step **Decides:** High-impact decisions about people need subgroup evaluation and oversight.

A	Dropped element	An overall score can hide poor results for a subgroup.
B	Camouflage	Vendor location does not address fairness or oversight.
C	Correct	Proxies can carry bias; subgroup testing and oversight are core responsible-AI controls.
D	Absolute language	Other inputs such as ZIP code can act as proxies.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	4	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	AI engineering vs. data science vs. research	AI engineering builds, deploys, and operates reliable AI systems; data science extracts insight from data; research invents new methods.	Treating a high notebook accuracy score as a finished engineering deliverable.	The question asks who owns uptime, deployment, or maintenance versus who answers a business question or publishes a new method.
2	Algorithmic complexity (Big-O)	Big-O describes how run time grows with input size: $O(n)$ scales linearly, $O(n^2)$ scales with the square of the growth factor.	Assuming every algorithm's time grows in step with the data.	A run time at one input size and a question about a larger size.
3	Train, validation, and test splits	The test set is touched once, at the end; tuning on it or leaking its information into training inflates reported performance.	Reusing the test set to pick hyperparameters and still calling its score unbiased.	A score that looks great offline and collapses in production, or a split made after preprocessing on all data.
4	Overfitting vs. underfitting	High training score with much lower validation score = overfitting (variance); low scores on both = underfitting (bias).	Fixing an overfit model by making it bigger or training longer.	Two scores side by side: training vs. validation.
5	Precision vs. recall	Precision = $TP / (TP + FP)$, the share of flagged cases that are real; recall = $TP / (TP + FN)$, the share of real cases that were caught.	Swapping the denominators.	Counts of cases flagged and cases that actually exist.
6	Latency vs. throughput	Latency is time per request; throughput is requests completed per unit time; batching can raise throughput while raising latency.	Assuming a change that improves throughput always improves user-facing response time.	Requests per second next to milliseconds per request.
7	Reliability and graceful degradation	A production AI system needs monitoring, drift detection, and a safe fallback when the model or its inputs fail.	Assuming a model that passed testing will keep performing without monitoring.	Input data changing over time, or a model service that times out.
8	Responsible AI practice	Evaluate performance across affected subgroups, document limits, and keep meaningful human oversight for high-impact decisions.	Believing that removing a protected attribute from the inputs guarantees a fair model.	A high-stakes decision about people and an overall accuracy number.

Course scope (public description): This course provides a foundational orientation to the field of AI engineering, focusing on the essential interplay between programming, mathematics, and real-world applications. Learners apply core programming principles and foundational mathematical ideas that support algorithmic thinking and analyze how these foundations enable the design and development of AI systems. Students also describe how AI engineering differs from related domains, such as data science and academic research, with an

emphasis on applied engineering practice. Through conceptual frameworks, applied examples, and introductory analysis, students begin to understand the skills, mindset, and responsibilities of AI engineers, including considerations related to scalability, reliability, and responsible use of AI technologies. The course establishes a strong foundation for success in the AI Engineering program by clarifying expectations for rigor and by highlighting the relevance of foundational competencies for building trustworthy, impactful, and real-world AI solutions.